

UNIT - 2 SDG

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Question 9: Represent a neural network architecture (input, hidden, output layers) suitable for predicting crop yield from soil moisture, temperature, and rainfall data. Justify the number of neurons chosen.

SDG 2 – Zero Hunger

Problem Statement

Predict crop yield using environmental factors such as soil moisture, temperature, and rainfall to help farmers improve agricultural productivity.

Background

Crop yield depends on several environmental conditions. Neural Networks can learn complex relationships between these factors and accurately predict crop production.

Input Features

- Soil Moisture
- Temperature
- Rainfall

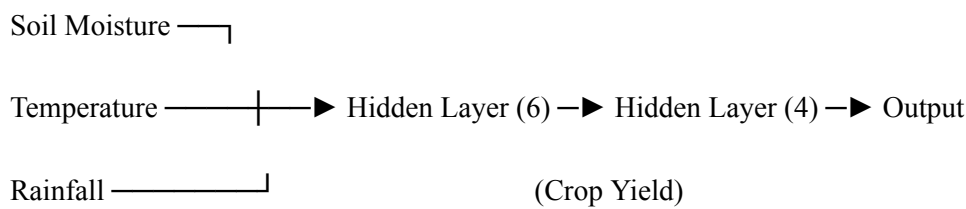
Output

- Predicted Crop Yield (High / Medium / Low or Yield in tons/hectare)

Neural Network Architecture

- **Input Layer:** 3 neurons (one for each input feature)
- **Hidden Layer 1:** 6 neurons with ReLU activation
- **Hidden Layer 2:** 4 neurons with ReLU activation
- **Output Layer:** 1 neuron with Linear activation (for yield prediction)

Architecture Diagram



Justification of Neurons

- **3 input neurons** represent the three environmental parameters.
- **6 neurons** in the first hidden layer learn complex relationships.
- **4 neurons** in the second hidden layer refine the learned patterns.
- **1 output neuron** predicts the crop yield value.

Advantages

- Learns nonlinear relationships.
- Provides accurate crop yield prediction.
- Helps farmers plan irrigation and fertilizer usage.

Limitations

- Requires sufficient historical data.
- Model accuracy depends on data quality.

Conclusion

A multilayer neural network effectively predicts crop yield, enabling better agricultural planning and supporting **SDG 2 – Zero Hunger**.

Question 31: Formulate a genetic algorithm (chromosome encoding, fitness function, selection method) to optimize the placement of solar panels in a smart city for maximum energy yield.

SDG 7 – Affordable and Clean Energy

Problem Statement

Optimize the placement of solar panels in a smart city to maximize energy generation while minimizing installation costs.

Background

Solar energy production depends on sunlight, roof orientation, shading, and available space. A Genetic Algorithm (GA) searches for the best panel arrangement.

Chromosome Encoding

Each chromosome represents a solar panel layout.

Example:

[1 0 1 1 0 1 0 1]

- 1 = Solar panel installed
- 0 = No panel installed

Fitness Function

The fitness function evaluates each solution:

Fitness = Total Energy Generated – Installation Cost – Shading Loss

Higher fitness indicates a better solar panel arrangement.

Selection Method

Tournament Selection

- Randomly select a small group of chromosomes.
- Choose the chromosome with the highest fitness.
- Repeat until enough parents are selected.

Genetic Operators

- **Crossover:** Exchange sections of two parent chromosomes to create new layouts.
- **Mutation:** Randomly change 0 to 1 or 1 to 0 to introduce diversity.

Workflow

1. Generate an initial population.
2. Evaluate fitness.
3. Select parents.
4. Perform crossover.
5. Apply mutation.
6. Create a new generation.
7. Repeat until the best solution is found.

Advantages

- Maximizes solar energy production.
- Reduces energy cost.
- Finds near-optimal solutions efficiently.
- Suitable for large smart-city planning problems.

Limitations

- Computationally expensive.
- Requires careful parameter tuning.

Conclusion

A Genetic Algorithm efficiently optimizes solar panel placement, increasing renewable energy production and supporting **SDG 7 – Affordable and Clean Energy**.

